



Team Details

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EdgeRunners

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Problem Statement Addressed



DESCRIPTION / DETAILS

AI-Based Restoration of Degraded Images for Semiconductor Inspection-KLA

Build an AI model that takes a degraded, noisy low-resolution semiconductor inspection image and restores it to match the original high-resolution, clean ground truth image, as closely as possible, as fast as possible



Idea Description – Describe your Idea/Solution/Prototype

KEY CONCEPT & APPROACH

DAF-Restormer Backbone: Restormer enhanced with dynamic degradation prompting and FFT frequency filtering.

Degradation Encoder: Predicts a 64-dim prompt vector & spatial noise map without assuming fixed noise distributions.

Spectral Bottleneck: FFT magnitude gating filters periodic grain patterns while preserving exact phase geometry.

SOLUTION OVERVIEW

Unclipped Signal Mapping: Restores 128x128 noisy inputs (<0 or >1) directly to clean 256x256 targets ($[0, 1]$).

2-Stage Training: 10-ep Charbonnier+SSIM pre-training + 2-ep LPIPS fine-tuning (**22% LPIPS reduction: 0.2719 $\rightarrow$ 0.2122**).

High Performance: **34.3 img/sec (35.8 ms/img)** on T4 GPU; reaches **28.47 dB PSNR & 0.7712 SSIM** with 8-view TTA.



Proposed Solution

SOLUTION DETAILS

Architecture Breakdown:

Degradation Encoder: Extracts 64-dim prompt vector & 128x128 spatial noise map from high-pass input features.

Prompted Transformer: MDTA (channel-transposed attention) + GDFN blocks modulated by degradation prompts.

Spectral Bottleneck: Prompt-conditioned FFT magnitude gating filters periodic grain while preserving phase.

ICNR PixelShuffle Head: 2x upsampling with ICNR init to eliminate sub-pixel checkerboard artifacts.

Progressive Supervision: Auxiliary 128x128 clean-LR reconstruction head + uncertainty estimation.

Tech & Training Stack: PyTorch 2.x (AMP FP16), AdamW, Cosine LR Decay, Charbonnier + SSIM + LPIPS loss.



Innovation and Uniqueness

KEY INNOVATION

Dynamic Degradation Prompting: Predicts prompt vectors & spatial noise maps directly from inputs without static noise assumptions.

Phase-Preserving FFT Filtering: Operates in Fourier space to damp periodic grain noise while strictly preserving phase geometry.

LPIPS Perceptual Fine-Tuning: Suppresses micro-grain artifacts that disrupt downstream semiconductor defect inspection.

COMPETITIVE ADVANTAGE

22% LPIPS Error Reduction: Drops LPIPS from 0.2719 to 0.2122, eliminating persistent micro-grain artifacts.

Top Benchmark Accuracy: Reaches **28.47 dB PSNR & 0.7712 SSIM** (+5.19 dB PSNR over Bicubic baseline).

Ultra-Fast 1-View Speed: Operates at **34.3 img/sec (35.8 ms/img)** on Tesla T4 GPUs to fit real-time tool execution budgets.



Impact and Benefits

Primary Impact

Automated Wafer Defect Inspection: Converts severely degraded 128x128 noisy micrographs into clean 256x256 ground truth estimates, enabling accurate detection of sub-10nm nanoscale semiconductor defect features.

Elimination of Inspection Bottlenecks: Ultra-fast single-pass processing restores inspection image streams in real time, preventing wafer throughput delays in high-volume fab environments.

Domain & Noise Robustness: Handles complex combined speckle and Gaussian degradations with valid out-of-bounds input intensities (<0 or >1) without requiring manual pre-clipping.

Quantifiable Outcomes

+5.19 dB PSNR Quality Gain: Achieves **28.47 dB PSNR** compared to the 23.28 dB non-learning Bicubic baseline.

+41.7% Structural Similarity Gain: Reaches **0.7712 SSIM** recovering sharp geometric line edges.

22% Perceptual Grain Elimination: Lowers LPIPS error from **0.2719 to 0.2122**, removing persistent noise artifacts.

34.3 img/sec Real-Time Speed: Delivers **35.8 ms/image** latency on Tesla T4 GPUs **Lightweight Model Footprint: 4.17M parameters**



Technology & Feasibility/Methodology Used

IMPLEMENTATION STRATEGY

End-to-End Pipeline: Modular PyTorch framework mapping 128x128 noisy inputs to 256x256 clean targets via joint 2x super-resolution and denoising.

Compute Feasibility: AMP FP16 precision requiring only **~2.3 GB peak VRAM**, running in **35.8 ms/img on Tesla T4 GPUs** (sub-10ms on H100 GPUs).

Deployment Compliance: Verified CLI (infer.py) outputs unclipped float32 arrays strictly matching problem brief specifications.

Software Architecture

PyTorch 2.x, CUDA 12.8, PyTorch AMP (FP16)
NumPy, SciPy, PyWavelets, TorchVision

Hardware Components

NVIDIA Tesla T4 / RTX GPUs (Validation)
NVIDIA H100 GPU (Target Evaluation)
4.17M Parameters

Development Tools

Git & Git LFS (Model Weights & Data Tracking)
Pytest (Automated Contract Verification)
Python 3.12 (Pinned requirements.txt)



GitHub & Video Link



GitHub Repository

<https://github.com/IAM-MUKUND/image-restoration-project>



Prototype / Simulation Video

<https://www.loom.com/share/704349d0f76b4823861861da1f7d43cf>



Research and References



Research Background & Methodology

Channel-Transposed Self-Attention: Applies MDTA (Multi-DConv Head Transposed Attention) across feature channels rather than spatial pixels, keeping computational complexity linear with image resolution.

Sub-Pixel Convolution & ICNR Init: Uses ICNR weight initialization for PixelShuffle upsampling to eliminate cross-channel variance imbalance and sub-pixel checkerboard artifacts.

Spectral Domain Filtering: Employs Fourier transform (FFT) magnitude gating to filter high-frequency grain noise while strictly preserving phase geometry.



References & Citations

Ref 1: Zamir et al., "*Restormer: Efficient Transformer for High-Resolution Image Restoration*", CVPR 2022.
(<https://arxiv.org/abs/2111.09881>)

Ref 2: Aitken et al., "*Checkerboard Artifact Free Sub-Pixel Convolution*", arXiv:1707.02937, 2017.
(<https://arxiv.org/abs/1707.02937>)

Ref 3: Zhang et al., "*Designing a Practical Degradation Model for Deep Image Super-Resolution*", ICCV 2021.
(<https://arxiv.org/abs/2103.14006>)